

Applied Mechanics Workbook



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Preface

This workbook provides a series of tutorials derived from lecture notes, offering clear and focused guidance on essential topics. Students are expected to be comfortable with college-level mathematics and science and are strongly encouraged to consult reliable reference texts. For example, foundational and practical texts in mathematics and applied mechanics, including Bird (2021), Hannah & Hillier (1995), Russell et al. (2021), and Bolton (2021).

Chapter 1

Units, Vectors, and Scalar Quantities

1.1 SI Units

The book adopts the **Système International (SI)** of units. The base units most relevant to mechanics are:

Quantity	Unit	Symbol
Mass	kilogram	kg
Length	metre	m
Time	second	s
Force	newton	N
Energy / Work	joule	J
Power	watt	W
Pressure / Stress	pascal	Pa

Derived units follow from these. For example:

$$1 \text{ N} = 1 \text{ kg} \cdot \text{m}/\text{s}^2 \quad 1 \text{ Pa} = 1 \text{ N}/\text{m}^2 \quad 1 \text{ J} = 1 \text{ N} \cdot \text{m}$$

1.2 Scalar and Vector Quantities

Scalar quantities have magnitude only and obey ordinary arithmetic:

- Examples: mass, speed, energy, temperature, time

Vector quantities have both magnitude and direction and must be combined using vector algebra:

- Examples: force, velocity, acceleration, displacement, momentum

A vector is represented graphically as an arrow — its length denotes magnitude, its orientation denotes direction.

1.3 Vector Addition

1.3.1 Triangle Rule

Two vectors **A** and **B** are added tip-to-tail; the resultant **R** closes the triangle:

$$\mathbf{R} = \mathbf{A} + \mathbf{B}$$

1.3.2 Parallelogram Rule

Both vectors are drawn from the same point; the resultant is the diagonal of the parallelogram they form.

1.3.3 Polygon Rule

For more than two vectors, they are placed tip-to-tail in sequence; the resultant closes the polygon from the tail of the first to the tip of the last.

1.4 Vector Subtraction

Subtracting **B** from **A** is equivalent to adding the negative of **B**:

$$\mathbf{R} = \mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$$

1.5 Resolution of Vectors

Any vector **F** can be resolved into two perpendicular components. For a vector at angle θ to the horizontal:

$$F_x = F \cos \theta$$

$$F_y = F \sin \theta$$

This is the most widely used technique in the book — resolving all forces into horizontal and vertical components before applying equilibrium conditions.

1.6 Resultant of Several Vectors

To find the resultant of a system of vectors analytically:

1. Resolve each vector into x and y components
2. Sum all components in each direction:

$$\Sigma F_x = F_{1x} + F_{2x} + \cdots \quad \Sigma F_y = F_{1y} + F_{2y} + \cdots$$

3. Find the magnitude of the resultant:

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

4. Find the direction:

$$\theta = \arctan\left(\frac{\Sigma F_y}{\Sigma F_x}\right)$$

1.7 The Laws of Sines and Cosines

For cases involving triangles of vectors, these laws provide an analytical alternative to graphical methods.

Sine rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule:

$$c^2 = a^2 + b^2 - 2ab \cos C$$

1.8 Summary of Equations

Concept	Equation
Horizontal component	$F_x = F \cos \theta$
Vertical component	$F_y = F \sin \theta$
Resultant magnitude	$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$
Resultant direction	$\theta = \arctan(\Sigma F_y / \Sigma F_x)$
Cosine rule	$c^2 = a^2 + b^2 - 2ab \cos C$

1.9 Summary

The resolution-and-recombination technique — breaking every vector into perpendicular components, summing each direction separately, then reconstructing the resultant — is the single most important skill introduced in this chapter. It underpins equilibrium analysis, dynamics, stress transformation, and virtually every other topic in the book.

Chapter 2

Balancing

2.1 Newton's Laws of Motion

2.1.1 First Law — Inertia

A body remains at rest or in uniform straight-line motion unless acted upon by a net external force.

A body in motion travels in a straight line unless forced to change direction. This resistance to any change in motion is called **inertia**. The law defines what a force *is*: the only thing capable of altering a body's state of motion.

$$\sum \vec{F} = 0 \implies \vec{v} = \text{constant}$$

2.1.2 Second Law — Force and Acceleration

The net force acting on a body equals its mass multiplied by its acceleration.

$$\boxed{F = ma}$$

Force and acceleration act in the same direction. The relationship is linear: doubling the net force doubles the acceleration; doubling the mass halves it for the same force.

Change	Effect on acceleration
Force doubled	Acceleration doubled
Mass doubled	Acceleration halved
Both doubled	Acceleration unchanged

2.1.3 Third Law — Action and Reaction

For every force one body exerts on another, the second body exerts an equal and opposite force back on the first.

$$\vec{F}_{A \rightarrow B} = -\vec{F}_{B \rightarrow A}$$

Key points:

- The forces are **equal in magnitude** and **opposite in direction**.
- They act on **different objects** — they are never a cancelling pair.
- The reaction force has the same nature as the action (both contact, both gravitational, etc.).

2.2 Centrifugal and Centripetal Forces

A body in motion travels in a straight line unless forced to change direction (Newton’s First Law). When moving in a circle, its inertia causes it to resist that continuous change of direction — in a rotating reference frame, this resistance appears as a fictitious outward “centrifugal” force. The real inward force that continuously deflects the body onto the circular path is the **centripetal force**. Because circular motion involves a constant change of direction, velocity is always changing even at constant speed. For small angles, the magnitude of this change in velocity simplifies to $v\theta$.

2.3 Static Balance

A system is statically balanced when the resultant of all centrifugal forces due to rotating masses is zero — i.e., the centre of mass lies on the axis of rotation. Static balance requires the vector sum of all mr terms to be zero:

$$\sum mr = 0$$

A single correction mass placed in the same plane can achieve static balance.

2.4 Dynamic Balance

Static balance alone is insufficient for systems with masses distributed across different axial planes. Dynamic balance requires two conditions to be satisfied simultaneously:

1. The vector sum of all centrifugal forces equals zero (static condition).
2. The vector sum of all **moments** of these forces about any reference plane equals zero.

$$\sum mr = 0 \quad \text{and} \quad \sum mr\ell = 0$$

This generally requires **two correction masses** placed in two chosen planes.

2.5 Balancing in a Single Plane

For coplanar rotating masses, a graphical or analytical approach uses the **force polygon**. Closing the mr vector polygon determines the required balancing mass and its angular position.

2.6 Balancing in Several Planes

For masses distributed along a shaft, the procedure is:

1. Choose a reference plane and construct a **couple polygon** using $mr\ell$ terms, where ℓ is the axial distance from the reference plane.
2. Closing the couple polygon gives the first correction mass (magnitude and angle).
3. Construct the **force polygon** using mr terms to find the second correction mass.

2.7 Balancing of Reciprocating Machinery

A reciprocating engine generates forces that are fundamentally harder to balance than rotating masses. The piston moves in one plane only — adding an opposite mass in that plane creates a new out-of-balance force perpendicular to it. The problem cannot be solved by a single counterweight alone.

2.7.1 Primary and Secondary Forces

Treating the connecting rod as two lumped masses — one at the crank pin, one at the gudgeon pin — splits the system into a rotating part and a pure reciprocating part. The total reciprocating force is:

$$F = mr\omega^2 \left(\cos \theta + \frac{r}{\ell} \cos 2\theta + \dots \right)$$

Term	Name	Frequency
$\cos \theta$	Primary force	ω (crank speed)
$\frac{r}{\ell} \cos 2\theta$	Secondary force	2ω
Higher terms	Negligible in practice	—

The secondary force arises from the obliquity of the connecting rod; higher harmonics are generally ignored. Written separately:

Primary force (first-order):

$$F_{\text{primary}} = mr\omega^2 \cos \theta$$

Secondary force (second-order):

$$F_{\text{secondary}} = \frac{mr\omega^2}{n} \cos 2\theta$$

where $n = \ell/r$ is the ratio of connecting rod length ℓ to crank radius r .

2.8 Methods for Complete Balancing

2.8.1 Crankshaft Counterweight (Partial Balance)

Adding a mass opposite the crank pin cancels the rotating component entirely but only reduces the primary reciprocating force — it cannot eliminate it. This trades a vertical unbalance for a smaller horizontal one.

2.8.2 Multi-Cylinder In-Line Engines

By phasing cranks at equal angles, primary and secondary forces and couples can cancel across the engine. Balance requires:

$$\sum mr = 0 \quad \text{and} \quad \sum mrl = 0$$

for both primary and secondary components.

In multi-cylinder engine balancing there are four things to check:

- **F (Force)**: the vector sum of all mr terms. If non-zero, there is a net radial force shaking the engine.
- **C (Couple)** — the vector sum of all mrl terms. If non-zero, there is a rocking moment twisting the engine about a transverse axis, even if the net force is zero.

Configuration	Primary F	Primary C	Secondary F	Secondary C
2-cyl, 180°	✓	✗	✓	✓
4-cyl, 180°	✓	✓	✓	✗
6-cyl, 60°	✓	✓	✓	✓

The **six-cylinder in-line engine** achieves complete balance for both primary and secondary effects.

2.8.3 Partial Balance

Complete balance is not always possible or practical — particularly for single-cylinder engines. In such cases a balance mass is added to reduce the dominant unbalanced force, and a residual force in one direction is accepted to eliminate the more harmful force in the perpendicular direction.

2.9 Critical Speed

Every rotating shaft is also an elastic beam with natural frequencies of lateral vibration. The **critical speed** is the rotational speed at which the excitation frequency from residual unbalance equals a natural frequency, producing resonance.

For a shaft carrying a disc, using the static deflection δ_{st} under its own load:

$$N_c = \frac{60}{2\pi} \sqrt{\frac{g}{\delta_{st}}}$$

This is convenient because δ_{st} can be directly measured or calculated from beam theory.

2.9.1 Behaviour Around Critical Speed

Speed regime	Shaft behaviour
Well below N_c	Shaft runs stiffly; centre of mass geometric centre
Approaching N_c	Amplitude grows rapidly — whirling begins
At N_c	Resonance; amplitude theoretically unbounded (damping limits it)
Well above N_c	Shaft self-centres; centre of mass migrates toward axis

Many turbines and high-speed spindles operate **supercritically** — above the first critical speed — and simply pass through it during run-up.

2.9.2 Multiple Critical Speeds

A real shaft has infinitely many lateral modes, each giving a critical speed:

$$N_{c_n} \propto n^2 \quad (\text{uniform simply-supported shaft})$$

The **first critical speed** is the most dangerous: it is encountered first and produces the largest deflection amplitude.

2.10 Summary of Equations

Concept	Equation
Centrifugal force	$F = mr\omega^2$
Static balance	$\sum mr = 0$
Dynamic balance	$\sum mr = 0$ and $\sum mr\ell = 0$
Primary reciprocating force	$F_{\text{primary}} = mr\omega^2 \cos \theta$
Secondary reciprocating force	$F_{\text{secondary}} = \frac{mr\omega^2}{n} \cos 2\theta$
Critical speed	$N_c = \frac{60}{2\pi} \sqrt{\frac{g}{\delta_{st}}}$

2.11 Summary

Balancing is fundamentally a **vector problem**. Both graphical (polygon) and analytical methods are used throughout. Dynamic balancing always requires satisfying both force *and* moment equilibrium, building progressively from simple single-plane cases up to the complexity of multi-cylinder engine analysis. Critical speed analysis adds a further dimension: even a perfectly balanced shaft can experience destructive resonance if its operating speed coincides with a lateral natural frequency.

References

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- Bolton, W. (2021). *Engineering science* (Seventh edition). Routledge.
- Hannah, J., & Hillier, M. J. (1995). *Applied mechanics* (3rd edition). Longman Pub Group.
- Russell, P. A., Jackson, L., & Embleton, W. (2021). *Applied mechanics for marine engineers* (7th edition). Reeds.

Attributions

This book incorporates figures, data, and other resources from the following sources:

- Figures in the formulae, the centroids and second moments of area, are adapted from the Wikipedia pages on Centroids and Second moments of area.

Colophon

This book was typeset with Quarto 1.9.36 using XeLaTeX and Latin Modern fonts, with Menlo for code blocks.

Technical Notes

The following tools and technologies were used in the preparation of this book:

- Python: Primary language for computations
- uv: Modern Python package manager and resolver
- Ruff: Fast Python linter and code formatter
- Black: Python code formatter
- Bash: Unix shell for task automation
- macOS: Main development operating system

Appendix A

SI System Common Mistakes

Using the SI system correctly is crucial for clear communication in science and engineering. Below are common mistakes in using the SI system, examples of incorrect usage, and how to correct them.

Table A.1: SI system rules and common mistakes

Concept	Mistake	Correct Usage	Notes
Use of SI Unit Symbols	m./s	m/s	Use the correct format without additional punctuation.
Spacing Between Value & Unit	10kg	10 kg	Always leave a space between the number and the unit symbol.
Incorrect Unit Symbols	sec, hrs, °K	s, h, K	Use the proper SI symbols; symbols are case-sensitive.
Abbreviations for Units	5 kilograms (kgs)	5 kilograms (kg)	Avoid informal abbreviations like “kgs”; adhere to standard symbols.

Concept	Mistake	Correct Usage	Notes
Multiple Units in Expressions	5 m/s/s, 5 kg/meter ²	5 m/s ² , 5 kg/m ²	Use compact, standardized formats for derived units.
Incorrect Use of Prefixes	0.0001 km	100 mm	Choose prefixes to keep numbers in the range (0.1 x < 1000).
Misplaced Unit Symbols	5/s, kg10	5 s ⁻¹ , 10 kg	Symbols must follow numerical values, not precede them.
Degrees Celsius vs. Kelvin	300°K	300 K	Kelvin is written without “degree”
Singular vs. Plural Units	5 kgs, 1 meters	5 kg, 1 meter	Symbols do not pluralize; full unit names follow grammar rules.
Capitalization of Symbols	Kg, S, Km, MA	kg, s, km, mA	Symbols are case-sensitive; use uppercase only where specified (e.g., N, Pa).
Capitalization of Unit Names	Newton, Pascal, Watt	newton, pascal, watt	Unit names are lowercase, even if derived from a person’s name, unless starting a sentence.
Prefix Capitalization	MilliMeter, MegaWatt	millimeter, megawatt	Prefixes are lowercase for (10 ⁻¹) to (10 ⁻⁹), uppercase for (10 ⁶) and larger (except k for kilo).
Formatting in Reports	5, Temperature: 300	5 kg, Temperature: 300 K	Always specify units explicitly.

Appendix B

Greek Letters

The following tables present the names of Greek letters and selected symbols commonly used in engineering courses, ensuring precise reference and avoiding reliance on informal descriptors such as “squiggle.”

Table B.1: Greek letters.

Lower Case	Upper Case	Name
α	A	alpha
β	B	beta
γ	Γ	gamma
δ	Δ	delta
ϵ	E	epsilon
ζ	Z	zeta
η	E	eta
θ	Θ	theta
ι	I	iota
κ	K	kappa
λ	Λ	lambda
μ	M	mu
ν	N	nu
ξ	Ξ	xi
$\omicron$	O	omicron
π	Π	pi
ρ	P	rho
σ	Σ	sigma
τ	T	tau
υ	Υ	upsilon
ϕ	Φ	phi
χ	X	chi

Lower Case	Upper Case	Name
ψ	Ψ	psi
ω	Ω	omega

Table B.2: Commonly used symbols in engineering courses.

Symbol	Name	Use	Course
Δ	Delta	Change	Thermodynamics
Δ	Delta	Displacement	Naval Architecture
∇	Nabla	Volume	Naval Architecture
Σ	Sigma	Sum	Thermodynamics, Naval Architecture, Applied Mechanics
σ	Sigma	Stress	Thermodynamics, Applied Mechanics
ϵ	Epsilon	Modulus of elasticity	Thermodynamics, Applied Mechanics
η	Eta	Efficiency	Thermodynamics
μ	Mu	Friction	Thermodynamics, Applied Mechanics
ω	Omega	Angular velocity	Thermodynamics, Applied Mechanics
ρ	Rho	Density	Thermodynamics, Naval Architecture
τ	Tau	Torque	Thermodynamics, Applied Mechanics